

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

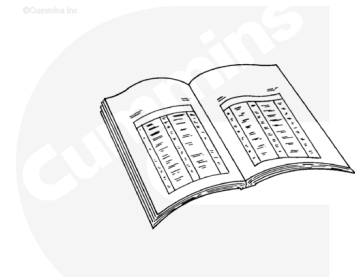
Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

- Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>).
- Remove the oil pan or oil pan covers. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html>), or Procedure Refer to Procedure 007-026 (</qs3/pubsys2/xml/en/procedures/56/56-007-026-tr.html>).
- Remove the oil pan adapter. Refer to Procedure 007-027 (</qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html>).

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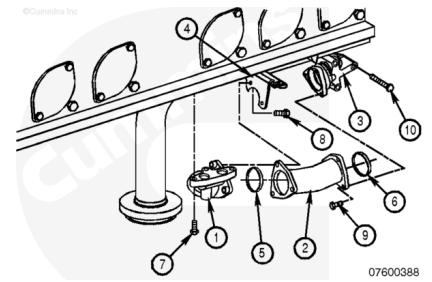


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Remove

S-Shape type

Note : Arrangement consists of lubricating oil pump connection (3) and block connection (1), with a steel tube (2) bridging the connections (1) and (3). Connection (1) has three mounting capscrews. Steel tube has slight bend in its section.



Remove the three capscrews (10) from the forward connector (3).

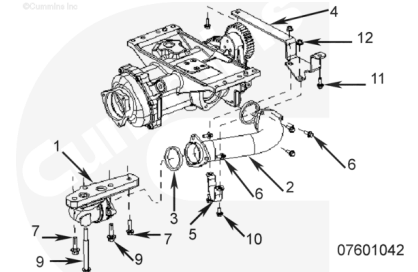
Remove the three capscrews (7) from the lubricating oil transfer tube to block connector (1).

Remove the two capscrews (8) from the bracket (4). Remove capscrews (9) from the lubricating oil transfer tube clamp and remove the lubricating oil transfer tube.

Remove remaining capscrews (9) from the lubricating oil transfer tube and connections.

Elbow type

Note : Arrangement consists of block connection (1), with a steel tube (2) which bridges from block connections (1) directly to lubricating oil pump.



Remove the two capscrews (10) and nuts (12) from the brackets (4) and (5). Remove tube brace (5).

Remove the four capscrews (7), (8), and (9), from the block connector (1).

Remove the capscrews (6) from the lubricating oil transfer tube to lubricating oil pump.

Remove the remaining capscrews (6) from the oil transfer tube (2) and connection (1).

Remove capscrews (11) and remove bracket (4).

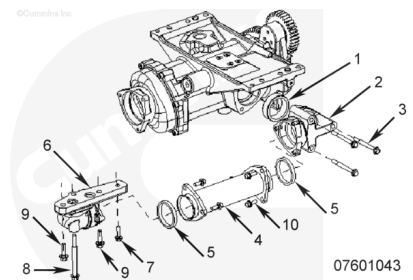
Straight type

Note : Arrangement consists of lubricating oil pump connection (2) and block connection (6), with a steel tube (4) bridging the connections (2) and (6). Connection (6) has four mounting capscrews. Steel tube is straight in section.

Remove the three capscrews (3) from the forward connector (2).

Remove the four capscrews (7), (8), and (9), from the block connector (6).

Remove the capscrews (10) from the lubricating oil transfer clamp tube (4) and remove the lubricating oil transfer tube.



Clean and Inspect for Reuse

⚠ WARNING ⚠

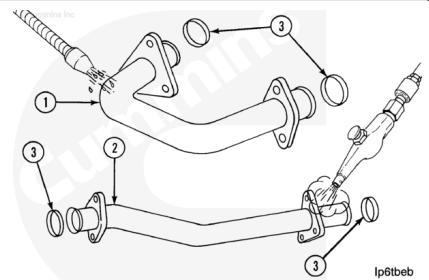
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



The oil transfer tubes for the QSK45 and QSK60 engines are different. The procedure is the same for both lubricating oil transfer tubes.

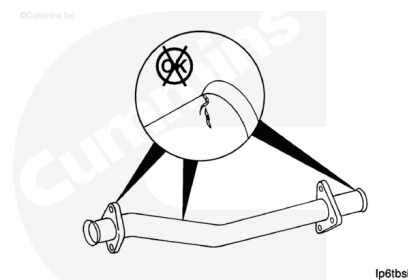
Remove the seals (3) from the tube and discard them.

Use solvent or steam to clean the lubricating oil transfer tube and dry with compressed air.

⚠ CAUTION ⚠

If the lubricating oil transfer tube is repaired by welding, be sure all of the welding slag and any metal that can break are removed the inside. If any debris enters the lubricating oil

pump and system, the gears will be damaged and failure will result.

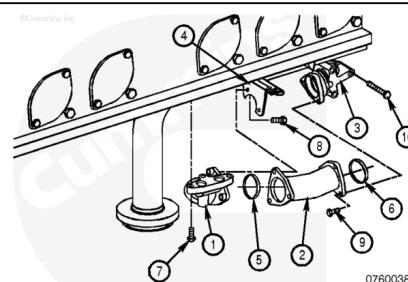


Check the lubricating oil transfer tube for cracks or other damage. If the lubricating oil transfer tube is cracked or damaged it **must** be repaired or replaced.

Install

S-Shape type

Note : Arrangement consists of lubricating oil pump connection (3) and block connection (1), with a steel tube (2) bridging the connections (1) and (3). Connection (1) has three mounting capscrews. Steel tube has slight bend in its section.



Lubricate the seals (5) and (6) with clean grease prior to installation.

Attach connector (1) to engine block with capscrews (10).

Install the lubricating oil transfer tube (2) with seal (5) with the three capscrews (9) into connector (1).

Attach connector (3) to the lubricating oil transfer tube (2) with seal (6).

Attach connector (3) with capscrews (10).

Tighten the capscrews except for the two bracket capscrews (8).

Torque Value:

M10 Capscrews

1. 45 n•m [33 ft-lb]

Torque Value:

M12 Capscrews

1. 80 n•m [59 ft-lb]

Loosen the capscrews that attaches bracket (4) to the cylinder block.

Note : Loosen the capscrews only enough to allow the bracket to move. The bracket **must** remain flat against the cylinder block.

Tighten the two capscrews (8) that fasten the bracket (4) and clamp to connector (1).

Tighten the two capscrews in the bracket to cylinder block connection.

Torque Value: 45 n•m [33 ft-lb]

Elbow type

Note : Arrangement consists of block connection (1), with a steel tube (2) which bridges from block connections (1) directly to lubricating oil pump.

Lubricate the seals (3) with clean grease prior to installation.

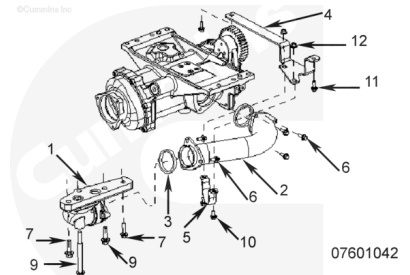
Install bracket (4) secure with two capscrews (11) to cylinder block.

Install oil transfer tube (2) to lubricating oil pump and secure with three off capscrews (6).

Install block connector (1) and secure with capscrews (7), (8), and (9) to cylinder block.

Install remaining capscrews (6) to transfer tube clamp (2) and block connector (1).

Install tube brace (5) and secure with capscrews (10) and nuts (12).



Torque Value:

M10 Capscrews

1. 45 n•m [33 ft-lb]

Torque Value:

M12 Capscrews

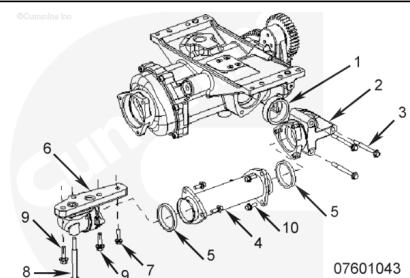
1. 80 n•m [59 ft-lb]

Straight type

Note : Arrangement consists of lubricating oil pump connection (2) and block connection (6), with a steel tube (4) bridging the connections (2) and (6). Connection (6) has four mounting capscrews. Steel tube is straight in section.

Lubricate seals (5) with clean grease prior to installation.

Install connector (6) with capscrews (7), (8), and (9).



Install the lubricating oil transfer tube (4) with seal (5), with three capscrews (10) into connector (6).

Install sleeve (1) into bore of lubricating oil pump.

Attach connector (2) to the lubricating oil transfer tube (4), with seal (5) and three capscrews (10).

Attache connector (2) with capscrews (3).

Torque Value:

M10 Capscrews

1. 45 n•m [33 ft-lb]

Torque Value:

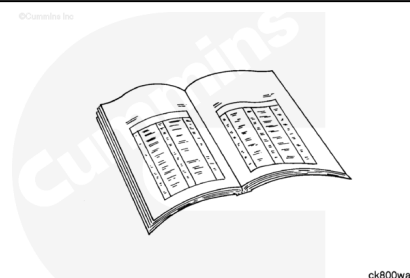
M12 Capscrews

1. 80 n•m [59 ft-lb]

Finishing Steps

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.



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- Install the oil pan adapter. Refer to Procedure 007-027 (/qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html).
- Install the oil pan or oil pan covers. Refer to Procedure 007-025 (/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html), or Procedure Refer to Procedure 007-026 (/qs3/pubsys2/xml/en/procedures/56/56-007-026-tr.html).
- Fill the lubricating oil. Refer to Procedure 007-037 (/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html).

